# AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

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| Project Title: Car Racing PyGame | |
| Project For: Mid Term Evaluation | Date of Submission: 28/03/2026 |
| Course Title: Python in Programming | |
| Course Code: **CSC 4262** | Section: D |
| Semester: Spring 25-26 | |

Group Name/No.: 8

| No | Name | ID | Program | Signature |
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| *Faculty use only* | | |
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| FACULTYCOMMENTS | **Marks Obtained** | |
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| | **Total Marks** | |

## 1. Project Overview & Track Selection

- **Track Name: Track 1 – Game Desktop App (GUI)**
- **Project Title:** Car Racing PyGame
- **Developer:** Md. Minhaj Rowfun Rabbi Anik
- **Date of Submission:** March 28, 2026

**Description:**

This project is a high-speed 2D racing game built using Python and the pygame-ce library. The application demonstrates core Python fundamentals and advanced Object-Oriented Programming (OOP) concepts. It features a lane-based navigation system, dynamic difficulty scaling and a persistent high-score csv as database.

At the heart of the experience is a Precision Lane-Based Navigation System. Unlike simple free-roaming movement, this system calculates exact X-coordinates for four distinct lanes, ensuring pixel-perfect alignment and collision detection. To provide an engaging user experience, the game features Dynamic Difficulty Scaling. Every 5 seconds, a "Level Up" event is triggered, non-linearly increasing obstacle velocity and reducing spawn intervals. This creates a progressive challenge that scales with the player's skill and survival time.

## 2. System Features

- **State-Driven Interface:** Includes distinct game states: Home Menu, High Scores leaderboard, Active Gameplay, and Game Over screens.
- **Dynamic Difficulty Scaling:** The game tracks time; every 5 seconds, the "Level" increases, which raises the speed and spawn rate of obstacles.
- **Advanced Asset Handling:** Utilizes a library of over 50 unique vehicle models, randomized for each obstacle spawn.
- **Data Persistence:** Implements a Top 5 High Score system that saves player data (Score, Level, and Top Score) locally.
- **Responsive Controls:** Precise lane-based movement optimized for desktop keyboard input.

**3. Screenshots:**

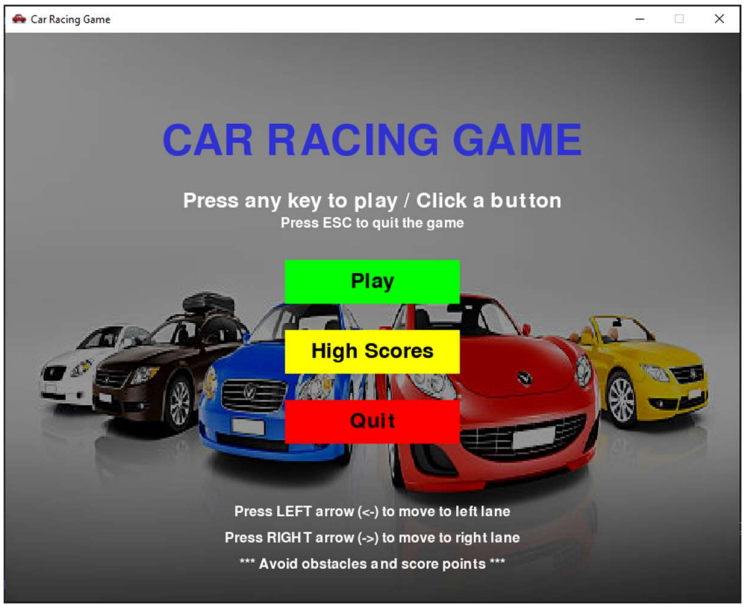



**Figure 1:** This is the Main Menu (Home Screen) of the game, serving as the central hub for the user interface. It utilizes a state-driven layout with interactive buttons to toggle between Gameplay, High Score records, and Exiting the application. Clear text overlays provide immediate instructions on player controls and objectives, ensuring an intuitive user experience.

**Figure 2.** This image represents the Active Gameplay state, showcasing the core mechanics of your 2D car racing project. It features a real-time Heads-Up Display (HUD) that tracks Score, Level and High Score, while demonstrating the lane-based movement system and randomized obstacle spawning logic.

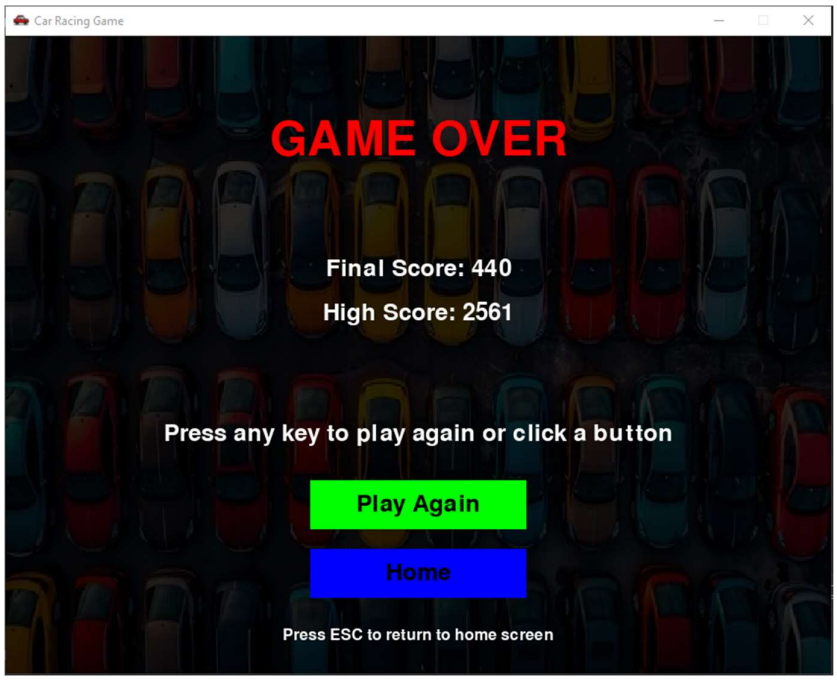



**Figure 3:** This image displays the Game Over state, triggered by a collision between the player and an obstacle. It highlights the session's Final Score compared to the all-time High Score, while providing clear navigation options to either restart the game immediately or return to the main menu.

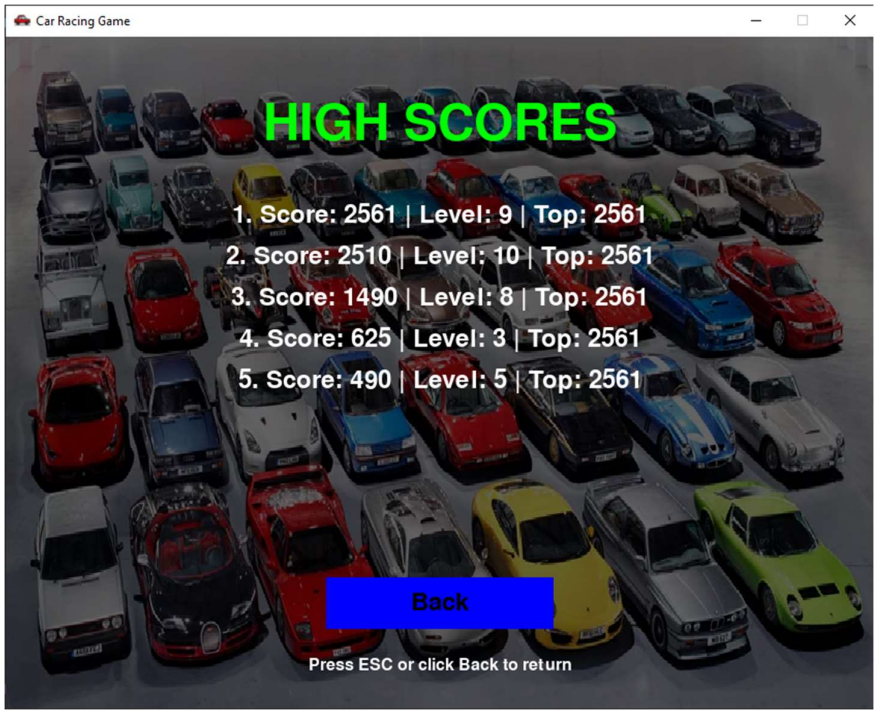
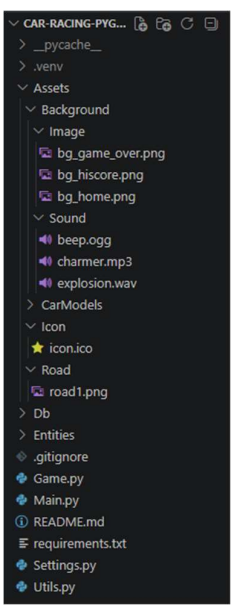
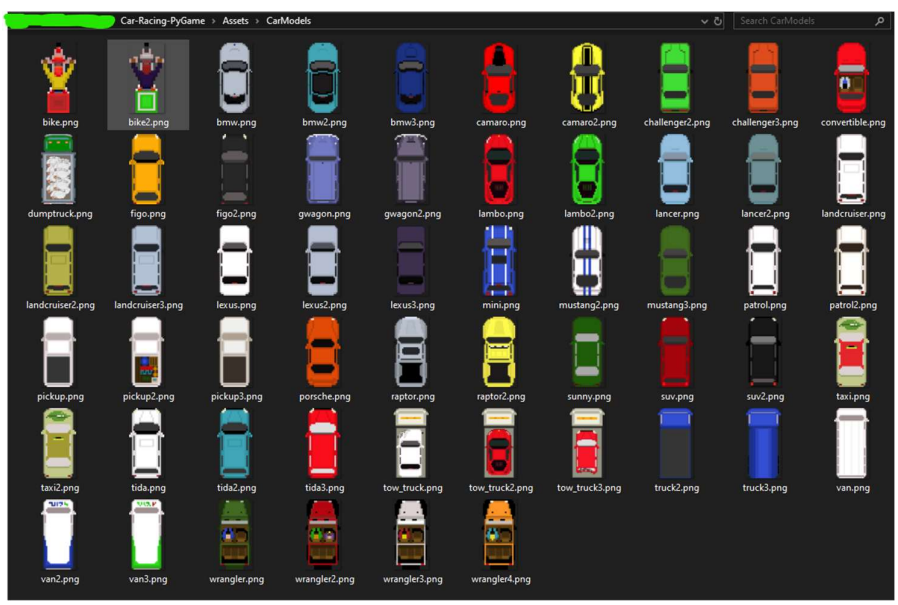



**Figure 4:** This image displays the High Scores leaderboard, fulfilling the project's requirement for a persistent summary report. It dynamically renders the top five records retrieved and sorted from the Hiscore.csv file, displaying each entry's specific score, level reached, and the all-time peak performance.

**3. Final Submission Structure:**

```
📂 Car-Racing-PyGame/
├── 📄 Main.py
├── 📄 Game.py
├── 📄 Settings.py
├── 📄 Utils.py
├── 📄 requirements.txt (contains: pygame-ce==2.5.7)
├── 📂 Entities/
│   ├── 📄 Car.py
│   ├── 📄 CarPlayer.py
│   └── 📄 CarObstacle.py
├── 📂 Db/
│   ├── 📄 FileLoader.py
│   ├── 📄 Paths.py
│   └── 📄 Hiscore.csv
├── 📂 Assets/ (Backgrounds, CarModels, Sounds, Icons)
└── 📄 Report.pdf
```




**6. Conclusion:**

The **Car Racing PyGame** project successfully integrates GUI development with advanced Python concepts. By decoupling physics, rendering and data storage into modular classes, the project provides a robust foundation for a desktop gaming application that meets all academic requirements for the Mid-term evaluation.